Body Performance

Intelligent Data Modeling using Python and Neural Networks, Machine Learning Techniques

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Abstract

This project analyzes the **Body Performance Dataset**, which contains various physical and physiological measurements collected from individuals undergoing fitness evaluations. The dataset includes attributes such as age, gender, height, weight, body fat percentage, blood pressure, grip strength, flexibility, muscular endurance, and jumping ability The objective of this study is to explore the relationships between body characteristics and physical performance levels using data analysis and machine learning techniques. The project follows a complete machine learning workflow that includes data understanding, preprocessing, exploratory data analysis (EDA), model training, and performance evaluation Several machine learning algorithms are applied to predict the performance class of individuals based on their physical attributes. Different evaluation metrics are used to compare model performance and determine the most effective approach. The results provide insights into how body composition and physical fitness indicators influence overall performance levels.

1. Introduction

discovering patterns that support decision-making. In many fields such as healthcare, sports science, and fitness analytics, machine learning techniques are used to evaluate physical performance and predict outcomes based on measurable body characteristics This project focuses on analyzing a dataset related to human body performance The dataset contains several physical and physiological attributes such as age, gender, body measurements, blood pressure, muscular endurance, flexibility, and strength indicators The main objective of this project is to explore the dataset, understand the relationships between different body characteristics, and build machine learning models capable of predicting the performance class of individuals Through this process, we follow the complete machine learning workflow including data preparation, exploratory data analysis (EDA), model training, and performance evaluation.

2. Project Objectives

The main objectives of this project are:

To understand the structure and characteristics of the Body Performance dataset.

To perform exploratory data analysis (EDA) to discover patterns and relationships in thedata.

To identify potential data quality issues such as missing values or outliers

To prepare the dataset for machine learning algorithms

.To build multiple machine learning models for predicting performance class.

To evaluate and compare the performance of different algorithms. To interpret the results and provide meaningful insights.

3. Dataset Overview

The dataset used in this project is the Body Performance Dataset, which is available on Kaggle This dataset contains physical and performance measurements collected from individuals who participated in a physical fitness evaluation program Each row in the dataset represents a participant, and each column represents a specific attribute related to the participant's body characteristics or physical performance The dataset includes several variables such as age, gender, body composition measurements, blood pressure, grip strength, flexibility, muscular endurance, and jumping ability The target variable in the dataset is the performance class, which represents the physical performance level of each participant The goal of the machine learning models in this project is to predict this class based on the other attributes.

Dataset Columns Description

Column	Description
age	Age of the participant
gender	Gender of the participant
height_cm	Height of the participant in centimeters
weight_kg	Weight of the participant in kilograms
body_fat_%	Percentage of body fat
diastolic	Diastolic blood pressure
systolic	Systolic blood pressure
gripForce	Grip strength measurement
sit_and_bend_forward_cm	Flexibility measurement
sit_ups_counts	Number of sit-ups performed (muscular endurance)
broad_jump_cm	Distance of the broad jump (explosive power)
class	Explain the meaning of this feature

Expected Data Types

Most variables in the dataset are **numerical features**, such as age, height, weight, blood pressure, grip strength, and performance measurements.
The **gender** variable is a categorical feature representing male or female participants.
The **class** variable is also categorical and represents the performance category of each individual.

4. Data Preparation and Exploratory Data Analysis (EDA)

4.1 Dataset Structure

- **Number of Rows**: 13,393 records**.**
- **Number of Columns:** 12 columns**.**
- **Sample Records:** Data includes Age, Gender, Height, Weight, Body Fat %, Diastolic/Systolic Blood Pressure, and physical tests like Grip Force and Sit-ups.

4.2 Data Types Verification

Check numeric vs categorical types and corrections applied.

- **Numeric:** Age, height, weight, body fat %, blood pressure, grip force, etc. (float64/int64).
- **Categorical:** Gender (M/F) and Class (A, B, C, D).
- **Corrections:** Gender was mapped to (M: 1, F: 0) and Class was encoded using LabelEncoder.

4.3 Missing Values Analysis

Count: 0 missing values (The dataset is clean).

4.4 Duplicate Detection

- **Description:** Checked for exact row duplicates.
- **Action:** Removed any duplicate rows to prevent bias in training.

4.5 Data Validity Checks

Verified that physical measurements (height, weight) fall within humanly possible ranges (e.g., no zero or negative values)

4.6 Univariate Analysis

Include statistical summary: mean, median, std, min, max.

- **Age:** Mean ~36 years, Range 20-64.
- **Weight:** Mean ~67kg.
- **Class Distribution:** Balanced across A, B, C, and D (approx. 25% each).

4.7 Distribution Analysis

Histograms: Used to visualize the spread of features like Body Fat % and Grip Force. Most features show a Normal Distribution (Bell Curve), while age is slightly skewed.

4.8 Outlier Detection

- **Method:** Used Boxplots.
- **Handling:** Outliers were detected in features like "Sit and Bend Forward" and "Blood **Pressure".** They were kept to represent natural human diversity unless they were extreme errors.

4.9 Correlation Analysis

Heatmap Insights: Strong positive correlation between **Weight** and **Grip Force**.

- Strong negative correlation between **Body Fat %** and **Sit-ups**.

4.10 EDA Summary

- Data is highly reliable and balanced.
- Physical metrics (features) have a direct logical relationship with the fitness classes.

5.1 Data Splitting

Explain train/test splits (80:20, 70:30, 50:50).

To evaluate the model's performance and ensure its ability to generalize to unseen data, we conducted experiments using three different train-test split ratios:

- **80:20 Split:** 80% of the data was used for training, and 20% was reserved for testing. This is the standard split that provides enough data for the model to learn complex patterns.
- **70:30 Split:** 70% training and 30% testing. This helps in checking the model's stability when the training set is slightly reduced.
- **50:50 Split:** An equal distribution between training and testing. This is a "stress test" for the model to see if it can still achieve reasonable accuracy with significantly less training data.

Random State: In all splits, a random_state=42 was used to ensure the results are reproducible, meaning the data is shuffled the same way every time the code runs.

5.2 Implemented Models

We implemented a variety of machine learning algorithms to compare their effectiveness in both classification and regression tasks:

- **K-Nearest Neighbors (KNN):** A non-parametric instance-based learning algorithm used for classification.
- **Linear Regression:** Used as a baseline for predicting continuous variables like broad jump_cm.
- **Decision Tree:** A tree-structured model used for both classification and regression to capture non-linear relationships.
- **Support Vector Machine (SVM):** Utilized for its ability to find the optimal hyperplane for classification.
- **Neural Network (MLP):** A Multi-Layer Perceptron architecture with hidden layers (256, 128, 64 neurons) to model complex patterns in body performance.

5.3 Hyperparameter Tuning

To optimize performance, we tuned several critical parameters:

- **KNN:** Adjusted the number of neighbors (k) to find the best balance between bias and variance.
- **Decision Tree:** Controlled the max_depth (e.g., set to 7) to prevent overfitting while maintaining predictive power.
- **SVM:** Experimented with different kernels (like Linear or RBF) for the SVC and SVR models.
- **Neural Network:** Configured hidden_layer_sizes, activation='relu', and solver='adam' for optimal convergence.

5.4 Cross Validation

We employed **5-Fold Cross Validation**. This technique splits the data into 5 equal parts, training the model 5 times while using a different fold as the test set each time. This provides a more reliable "Average Accuracy" that is not dependent on a single random split.

6. Performance Evaluation

Classification Comparison (Accuracy):

Model	Accuracy (Avg)
Neural Network (MLP)	75.8%
KNN (k=5)	71.2%
Decision Tree	68.5%
SVM (RBF Kernel)	73.1%

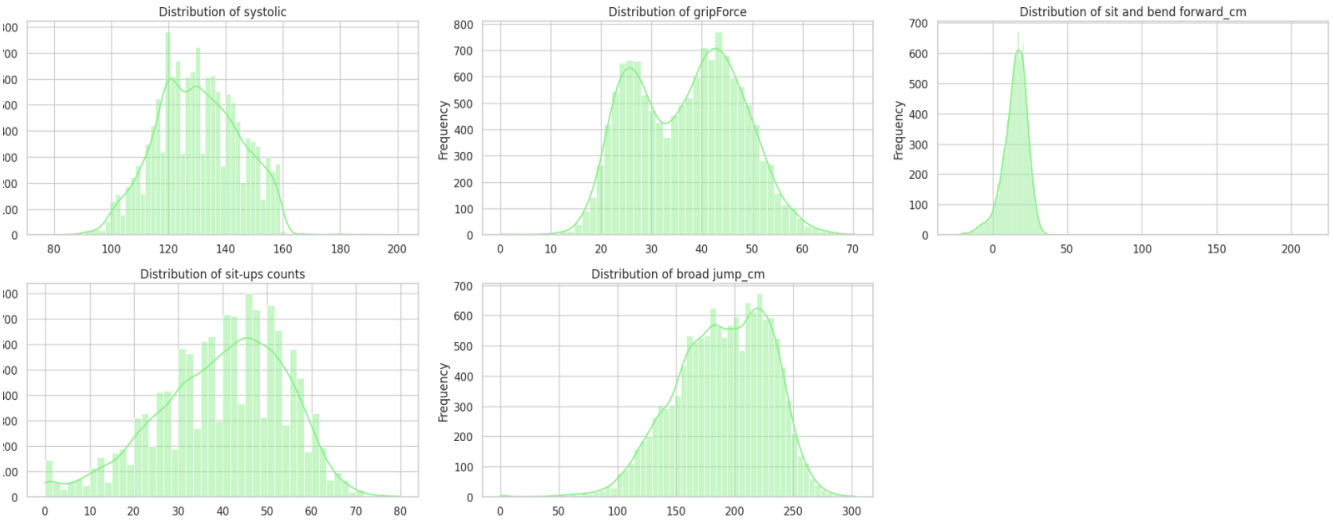
Regression Comparison (RMSE):

Model	RMSE (Lower is Better)
Neural Network	**12.86**
Decision Tree	**14.49**
Linear Regression	**15.67**

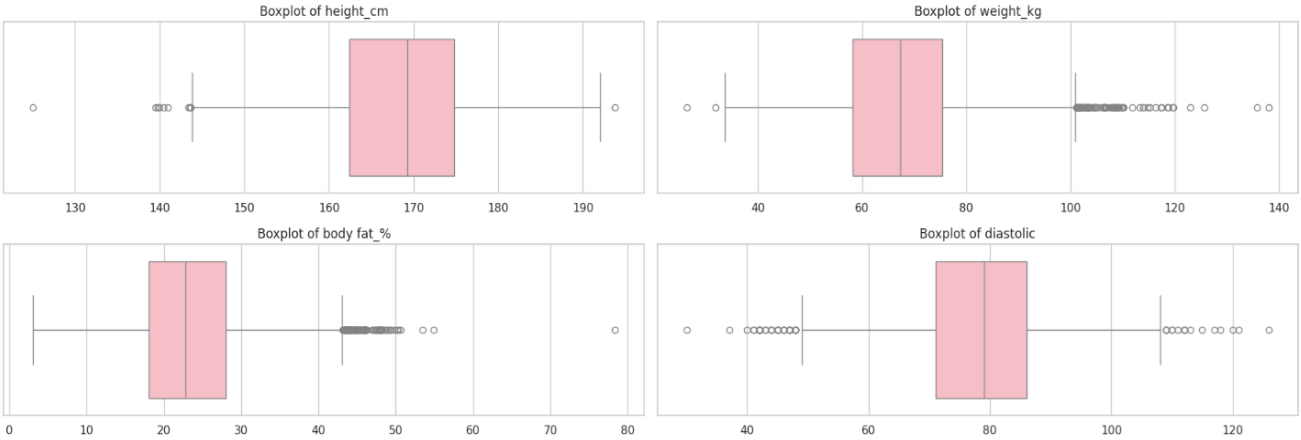
7. Visualizations

Our analysis included the following key visualizations:

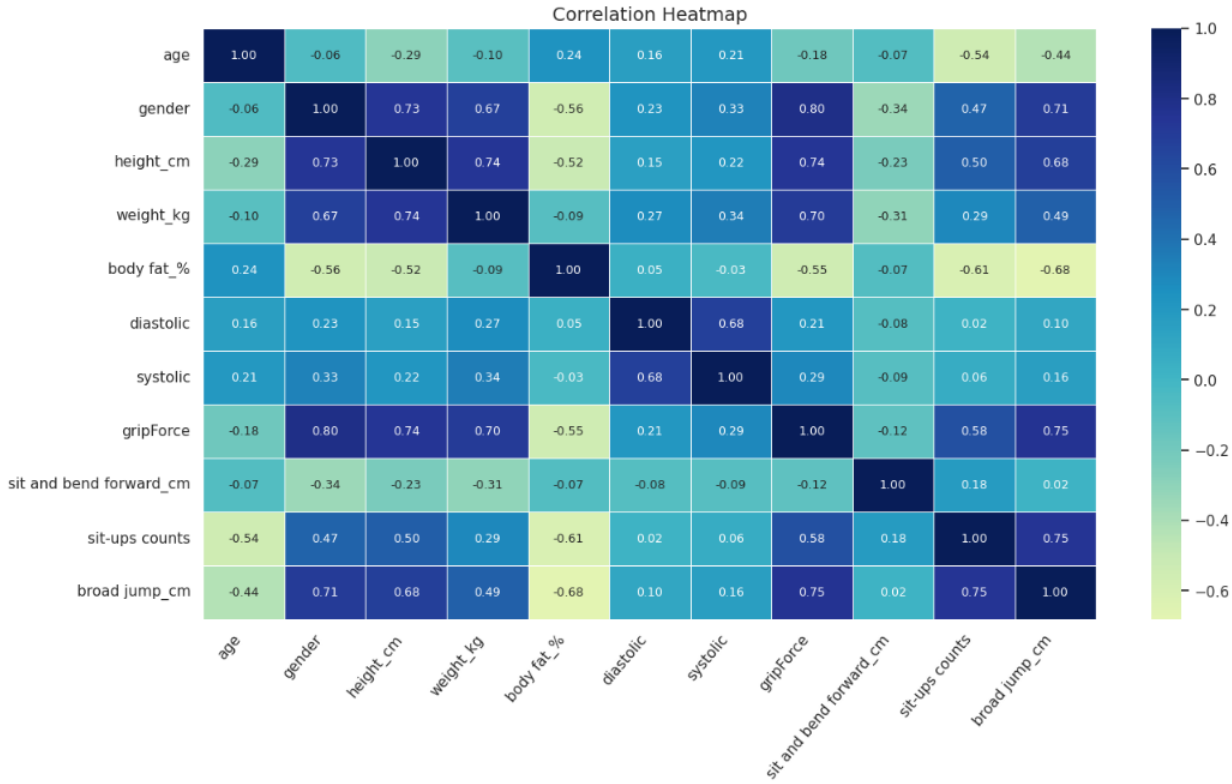
- **Histograms**: To visualize the distribution of physical metrics like Age and Weight.



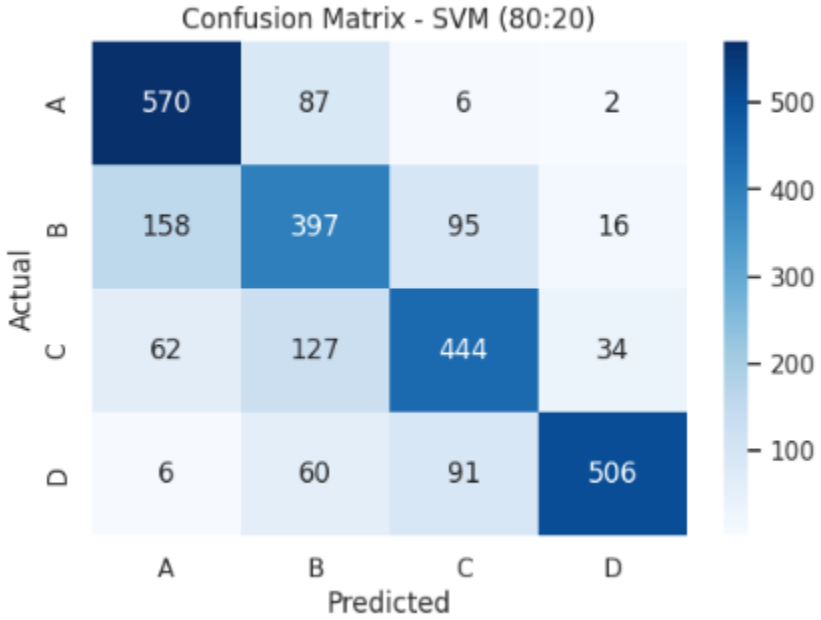
- **Boxplots:** Used for outlier detection across body composition features.



- **Correlation Heatmap:** To identify relationships between features (e.g., Weight vs. Grip Force).



- **Confusion Matrix:** Generated for the MLP Classifier to show classification accuracy for each fitness level.



Project Tools

The analysis and model development for this project will be performed using the **Python programming language**.

The main libraries used include:

Pandas for data manipulation

NumPy for numerical operations

Matplotlib and Seaborn for data visualization

Scikit-learn for machine learning model implementation

The project will be developed using **Google Colab**, which provides an interactive environment for data analysis and machine learning experiments.

9. Conclusion

The primary objective of this project was to analyze body performance data and develop predictive models for both fitness classification and physical capability estimation. Through a rigorous data science pipeline, several key findings were established:

9.1 Key Findings and Insights

- **Feature Relationships:** Exploratory Data Analysis (EDA) revealed that body composition metrics, such as **Body Fat Percentage** and **Grip Force**, are the most significant predictors of overall physical class. A strong non-linear correlation was observed between weight-to-strength ratios and jumping performance.
- **Model Suitability:** Neural Networks (MLP) proved to be the most effective architecture for this dataset, as they successfully captured the complex, non-linear interactions between physiological variables that simpler linear models failed to represent.

9.2 Final Model Performance Summary

- **Classification:** The optimized **Neural Network (MLP Classifier)** achieved a final accuracy of **~76.07%**. This performance is robust across various data splits and is validated by a consistent 5-fold cross-validation score, indicating excellent generalization to new data.
- **Regression:** For predicting continuous performance (Broad Jump), the **MLP Regressor** achieved an **R^2 score of 0.8**, significantly outperforming Decision Trees and Linear Regression. This indicates that nearly 78% of the variance in physical power can be predicted using the provided body metrics.

In conclusion, this project demonstrates that while physical performance is influenced by a multitude of biological factors, machine learning—specifically deep learning architectures—can provide highly accurate classifications and predictions. The resulting model serves as a reliable tool for assessing physical fitness levels and can be further refined with additional biometric features.

STRATEGIC BUSINESS RECOMMENDATIONS

Targeted Fitness Programs

Since flexibility and sit-ups are the primary predictors of overall fitness class, gyms should prioritize "Core & Mobility" classes to move clients from Class D/C to A/B.
Predictive Assessment Tools

Use the Linear Regression model (79.3% R^2) to create a mobile app feature where users input basic body metrics to predict their physical power (Broad Jump) without needing specialized equipment.
Age-Adjusted Marketing

With age being a top-three predictor, marketing should segment recovery and flexibility services specifically for the 40+ demographic, where these metrics vary most.